

PRACTICAL 1

Aim : Connect computer using given topology with wired media. Assume six devices are arranged, if in:

- a) Bus topology
- b) Ring topology
- c) Star topology
- d) Mesh topology

Find out number of cables (links), ports needed in each device and total number of ports needed in entire network for each of above stated topology.

Input :

a) Bus Topology –

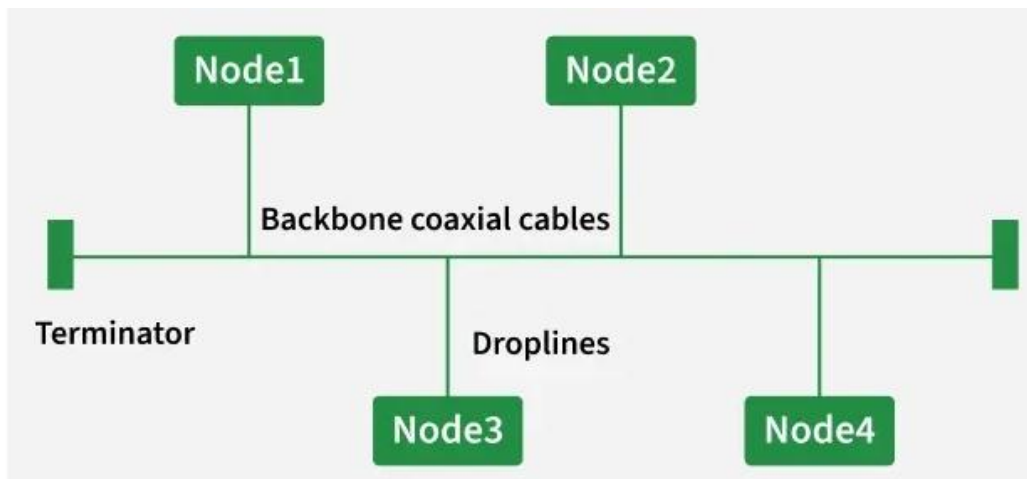
In a bus topology all device are connect to a single central cable/device and the data travel along the bus in both direction, each device only require one port to the bus.

Advantages

- If N devices are connected to each other in a bus topology, then the number of cables required to connect them is 1, known as backbone cable, and N drop lines are required.
- Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.
- The cost of the cable is less compared to other topologies, but it is used to build small networks.
- Bus topology is familiar technology as installation and troubleshooting techniques are well known.
- CSMA/CD was the only MAC method used in traditional bus Ethernet. Modern switched Ethernet does not use CSMA/CD because full-duplex operation eliminates collisions.

Disadvantages

- A bus topology is quite simpler, but still, it requires a lot of cabling.
- If the common cable fails, then the whole system will crash down.
- If the network traffic is heavy, it increases collisions in the network. To avoid this, various protocols are used in the MAC layer known as Pure Aloha, Slotted Aloha, CSMA/CD, etc.
- Adding new devices to the network would slow down networks.
- Security is very low.



b) Ring Topology –

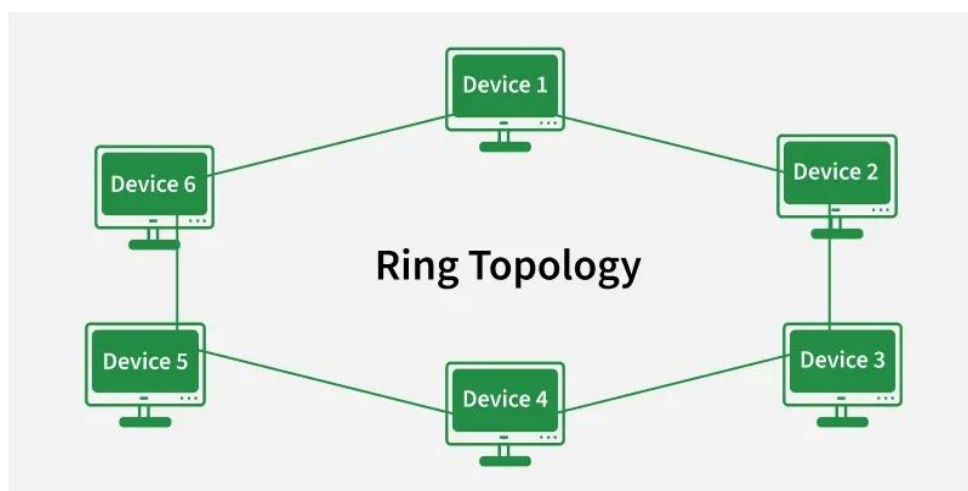
In a ring topology each device is connect to two other device, in a circular faction, so each device has two ports, for incoming data and one port for outgoing data. Data move in unidirectional and bidirectional.

Advantages

- The data transmission is high-speed.
- The possibility of collision is minimum in this type of topology.
- Cheap to install and expand.

Disadvantages

- The failure of a single node in the network can cause the entire network to fail.
- Troubleshooting is difficult in this topology.
- Less secure.



c) Star Topology –

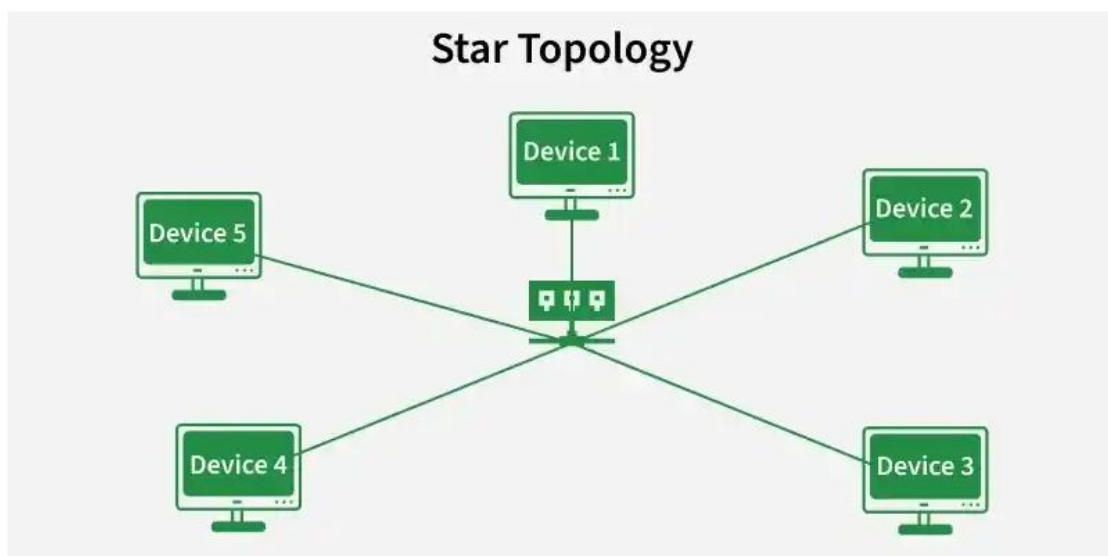
In a star topology each device is connected to a central device (hub/switch) using on cable, the central device manage communication between device.

Advantages

- If N devices are connected to each other in a star topology, then the number of cables required to connect them is N . So, it is easy to set up.
- Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N .
- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.
- Star topology is cost-effective as it uses inexpensive coaxial cable.

Disadvantages

- If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.
- The cost of installation is high.
- Performance is based on the single concentrator i.e. hub.



d) Mesh Topology –

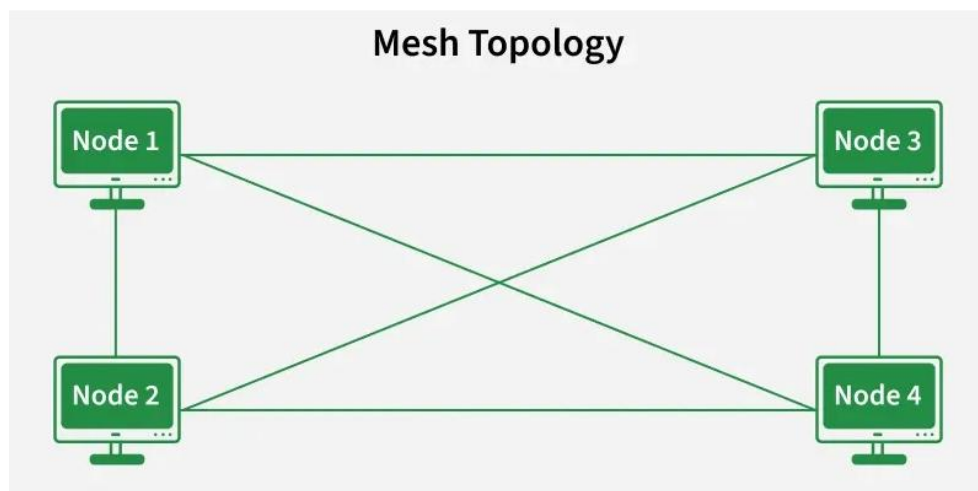
In a mesh topology every device is directly connect to every other device, in this case the number of connection increase significantly with number of device.

Advantages

- Communication is very fast between the nodes.
- Mesh Topology is robust.
- The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.
- Provides security and privacy.

Disadvantages

- Installation and configuration are difficult.
- The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.
- The cost of maintenance is high.



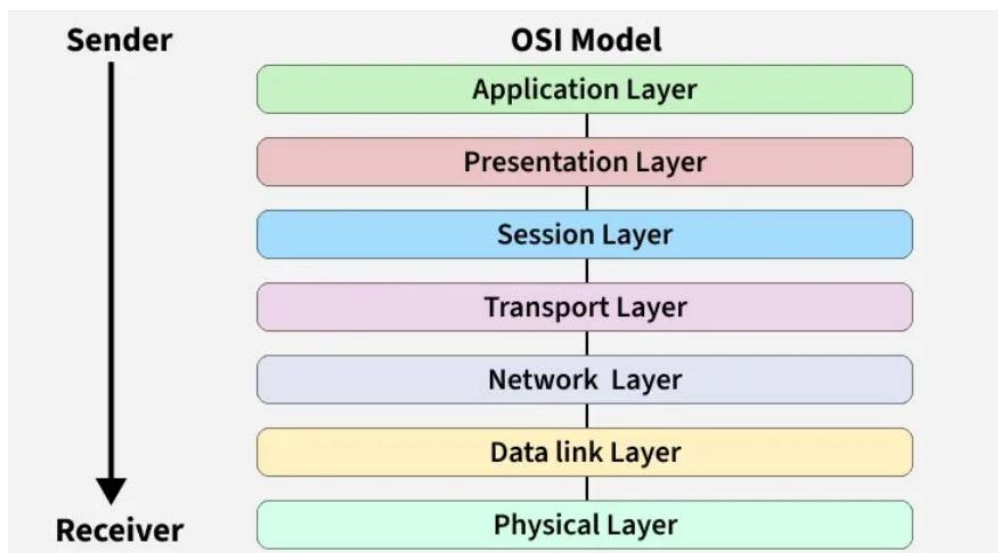
PRACTICAL 2

Aim : Study about OSI model network Layers.

Input :

The OSI Model is a conceptual framework created by the International Organization for Standardization (ISO) to describe how data is transmitted across a network using a structured seven-layer architecture.

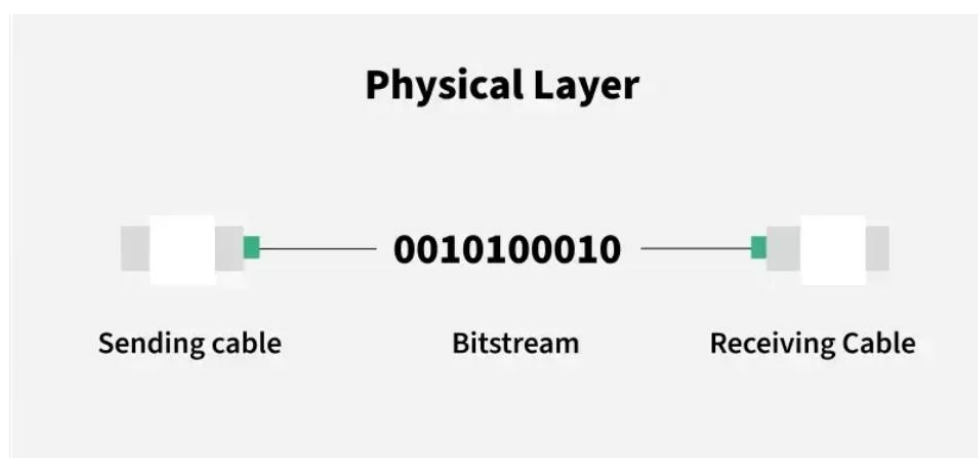
- Divides network communication into seven functional layers.
- Assigns specific responsibilities to each layer.
- Promotes compatibility between different networking systems.
- Simplifies network design, implementation, and troubleshooting.



Layer 1 : The Physical Layer

The Physical Layer is the foundation of the OSI model, acting as the bridge for actual physical connections between devices. Its primary mission is the transmission of raw, unstructured bitstreams over a physical medium from one node to the next.

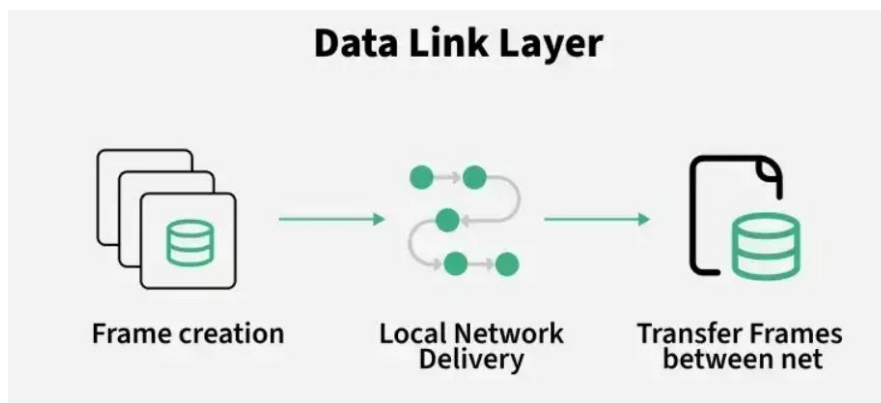
- **Core Responsibility** : It manages the hardware-level transmission of individual bits. When receiving data, it translates incoming physical signals back into digital 0s and 1s to be processed by the Data Link layer.
- **Essential Hardware** : Common devices operating at this level include Hubs, Repeaters, Modems, and Cables.
- **Bit Synchronization** : It ensures the sender and receiver are "in sync" by providing a clock signal that controls the timing of bit transmission.
- **Bit Rate Control** : It dictates the transmission speed, defined as the number of bits sent per second.
- **Physical Topologies** : It defines the structural layout of the network, such as Bus, Star, or Mesh configurations.
- **Transmission Mode** : It determines the direction of data flow, utilizing modes like Simplex (one-way), Half-Duplex (two-way, one at a time), or Full-Duplex (simultaneous two-way).



Layer 2 : The Data Link Layer (DLL)

The Data Link Layer serves as the bridge between the physical hardware and the logical network, ensuring reliable node-to-node delivery of data. Its primary goal is to ensure that data transfer is error-free across the physical medium.

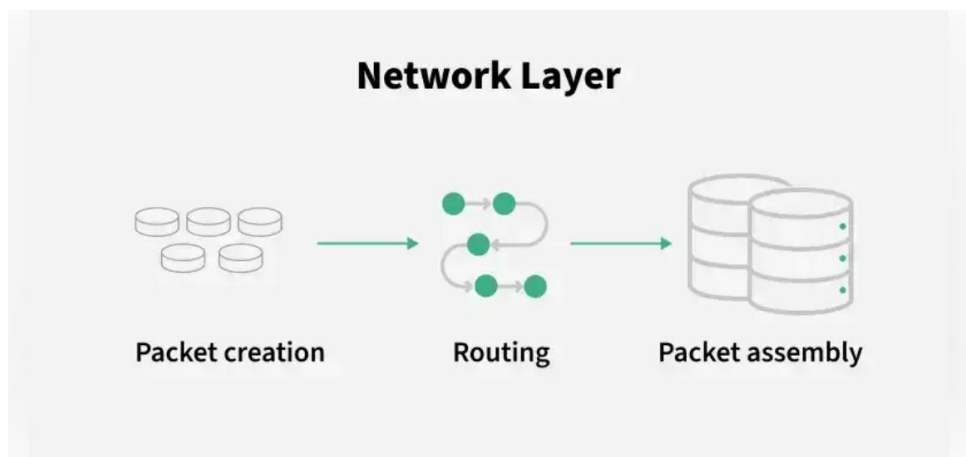
1. **Data Packaging (Framing) :** At this layer, data packets received from the Network layer are divided into manageable units called Frames. This is done by adding specific "start" and "stop" bits so the receiver can recognize where one unit of data ends and the next begins.
2. **Addressing and Hardware :** The DLL uses MAC addresses (Physical Addressing) to identify hosts. It encapsulates the sender's and receiver's MAC addresses into the frame header. Common devices operating here are Switches and Bridges.
3. **Sublayers :**
 - Logical Link Control (LLC): Manages flow and error control while acting as an interface for upper-layer protocols.
 - Media Access Control (MAC): Determines how devices physically access the transmission medium and handles hardware addressing.
4. **Error Control :** It detects and retransmits damaged or lost frames to maintain data integrity.
5. **Flow Control :** It synchronizes the data rate between a fast sender and a slower receiver to prevent data "bottlenecks" or loss.
6. **Access Control :** When multiple devices share the same communication channel, the MAC sublayer dictates which device has the right to transmit at any given time to avoid collisions.



Layer 3 : The Network Layer

The Network Layer manages data transmission between hosts across different networks by handling logical addressing and path finding.

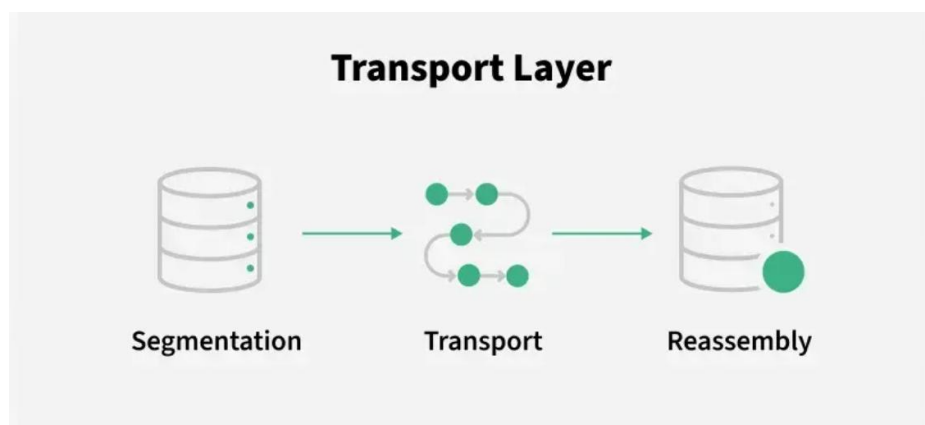
- **Data Unit** : Data segments are encapsulated into Packets.
- **Logical Addressing** : Assigns unique IP addresses (sender and receiver) to the packet header to identify devices globally.
- **Routing** : Determines the most efficient physical path from the source to the destination across interconnected networks.
- **Hardware** : Primarily implemented via Routers and Switches.
- **Inter-networking** : Facilitates communication between disparate networks by directing traffic to the correct destination.



Layer 4 : The Transport Layer

The Transport Layer ensures the end-to-end delivery of entire messages. It acts as a liaison, providing services to the Application layer while utilizing the infrastructure of the Network layer to ensure data reaches the correct application on the destination host.

- **Data Unit** : Data is broken down into Segments.
- **Service Point Addressing** : Uses Port Numbers (e.g., Port 80 for web traffic) to ensure data is delivered to the specific process or application intended, not just the device.
- **Segmentation & Reassembly** : Splits large messages into smaller segments for transmission and reassembles them in the correct order at the destination.
- **Protocols** : Common protocols include TCP (reliable), UDP (fast).
- **Connection-Oriented (TCP)** : Requires a "handshake" to establish a connection; ensures reliability via error checking and acknowledgements.
- **Connectionless (UDP)** : Sends data immediately without a formal connection; faster but offers no guarantee of delivery.



Layer 5 : The Session Layer

The Session Layer acts as the "dialogue manager," governing the opening, closing, and security of communication channels between two devices.

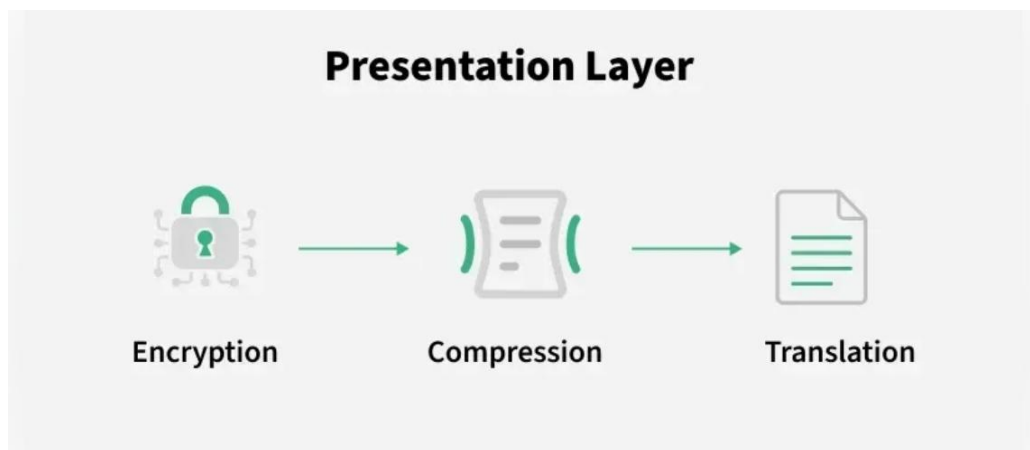
- **Session Lifecycle** : Manages the establishment, maintenance, and termination of connections between applications.
- **Authentication & Security** : Ensures that the communicating parties are verified and the connection is secure.
- **Synchronization** : Inserts checkpoints into the data stream. This allows a transfer to resume from the last saved point if a failure occurs, rather than restarting from the beginning.
- **Dialog Control** : Directs whether communication is half-duplex (alternating) or full-duplex (simultaneous).
- **Practical Example** : In a web-based messenger, the Session Layer maintains the active link between your browser and the server, ensuring your specific chat remains open and synchronized while handling background encryption and data conversion.



Layer 6 : The Presentation Layer

Often called the Translation Layer, the Presentation Layer ensures that data is formatted, secured, and compressed so that the receiving application can correctly interpret it.

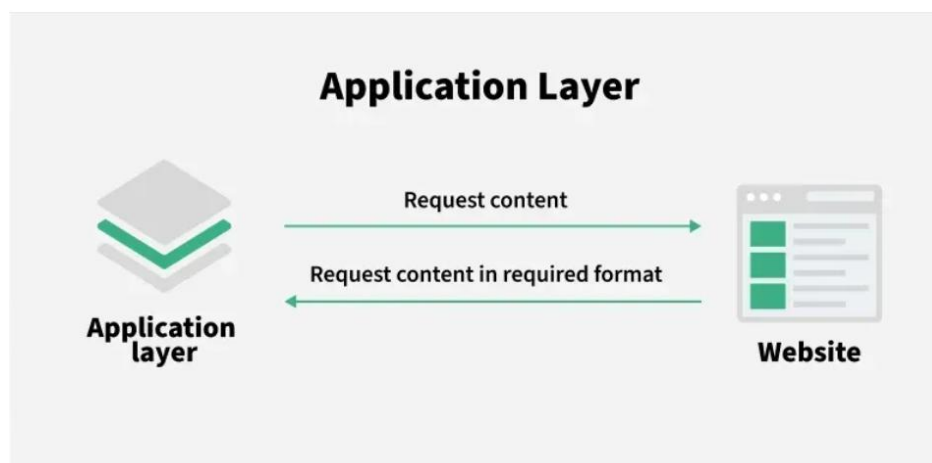
- **Translation** : It extracts data from the Application Layer and converts it into a standardized format for network transmission, bridging differences in data representation between different systems.
- **Standards & Formats** : Handles the encoding of media using standards such as JPEG, MPEG, and GIF.
- **Encryption/Decryption** : Provides security by converting "plain text" into "ciphertext" (encryption) and back again (decryption) using key values. This process is typically handled by protocols like TLS/SSL.
- **Compression** : Reduces the total number of bits required for transmission, which increases network efficiency and speed.



Layer 7 : The Application Layer

The Application Layer sits at the top of the OSI stack and serves as the direct interface between the software user and the network. It produces the data that will be sent and displays the information received from other layers.

- **User Interface** : Acts as a "window" for network-based applications (like web browsers or email clients) to access network services.
- **Core Protocols** : Utilizes high-level protocols such as HTTP/S (Web), SMTP (Email), FTP (File Transfer), and DNS (Domain Name Resolution).
- **Network Virtual Terminal (NVT)** : Enables users to log into and interact with remote hosts as if they were physically present at the terminal.
- **File Transfer, Access, and Management (FTAM)** : Provides the framework for users to retrieve, manage, and manipulate files stored on remote computers.
- **Directory Services** : Offers distributed database access to manage global information regarding various network objects and services.



PRACTICAL 3

Aim : Prepare and Test Straight UTP Cable and Cross UTP Cable.

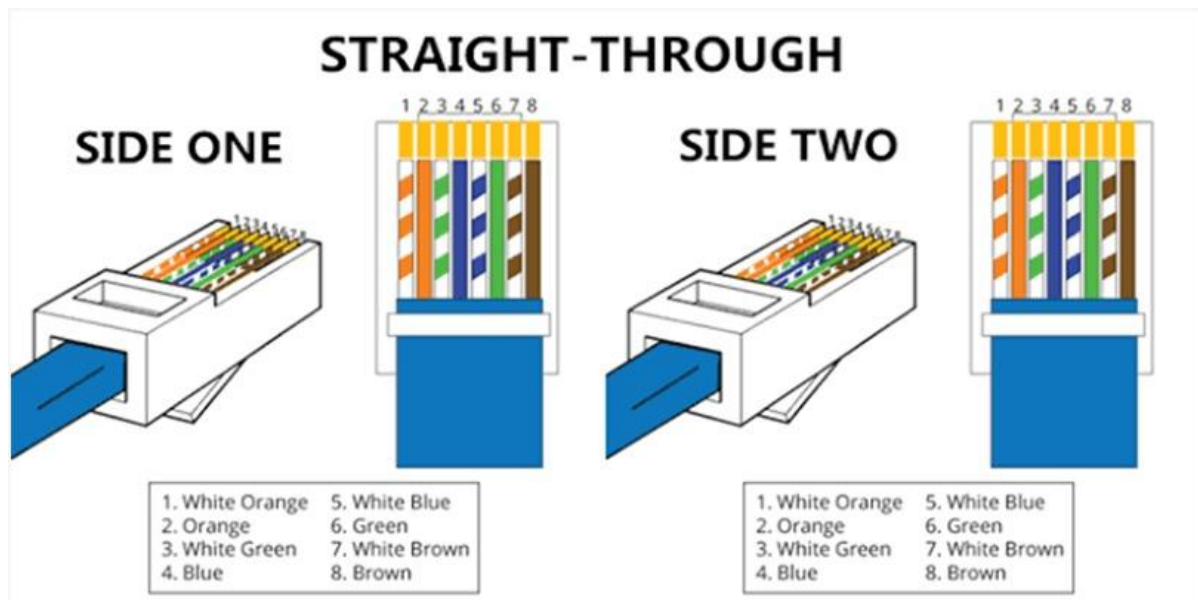
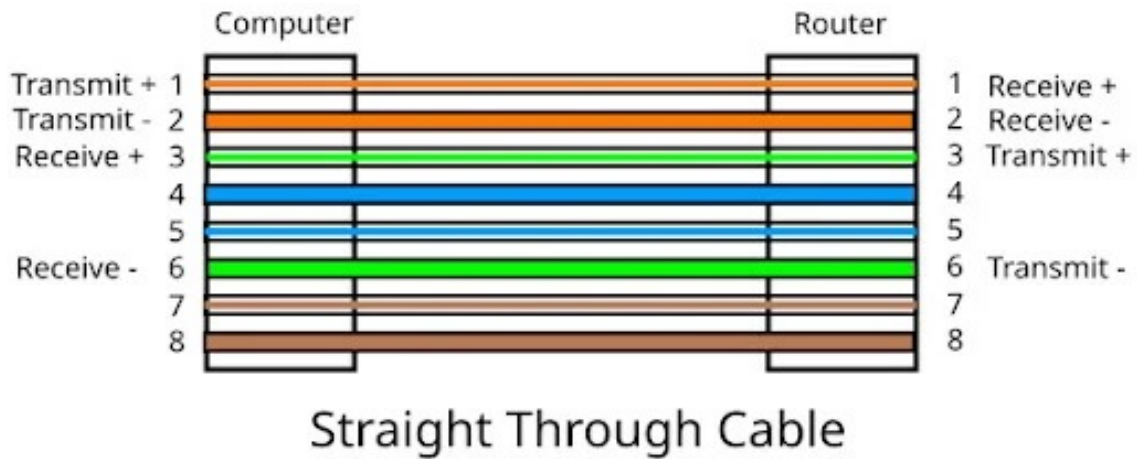
Input :

What is Straight Through Cable?

- A straight-through cable is a type of Ethernet cable used to connect different types of devices within a network.
- When connecting between two dissimilar devices, for example, from the computer to the router, a straight-through cable will be needed. There are no left or right constrictions on the straight-through cable, as both ends have the same wiring signals, so the wires correspond to the same pins at the other end, too. In such a way, the transceiver protocol can be set up in such a manner that the information is transmitted from one device to another without any need for manual configuration.

Features of Straight-Through Cable -

- **Direct Connection :** Links devices of different categories for instance a computer with a switch or a router, Computer-to-Device connection.
- **Standard Ethernet Cable :** It is particularly used in Ethernet networks for data transmission purposes.
- **Color Code Consistency :** The endings of the electrodes have identical sequences in terms of the color codes applied to the wires.
- **Common Usage :** Sometimes used in home and office networks for sharing devices among the nodes and link points.

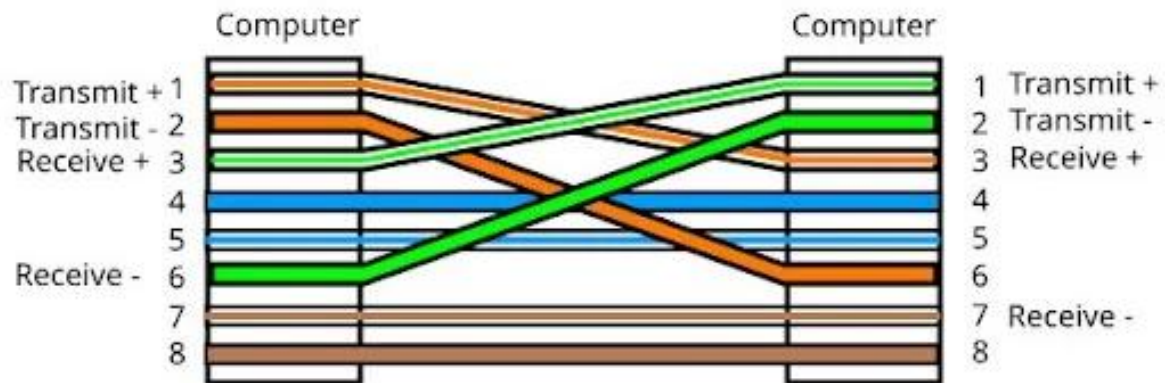


What is a Crossover Cable?

- A crossover cable is a type of Ethernet cable used to connect similar devices directly to each other without needing an intermediate network device like a switch or hub.
- Crossover cables are found primarily on connections between similar devices like two computers or two switches. A difference between straight-through and crossover cables is that the order of the wires differs at each end of the cable. This will allow communication of devices indicating if they are transmitting or receiving by 'crossover'ing the 'Tx' and 'Rx' so that data can be obtained by both similar devices.

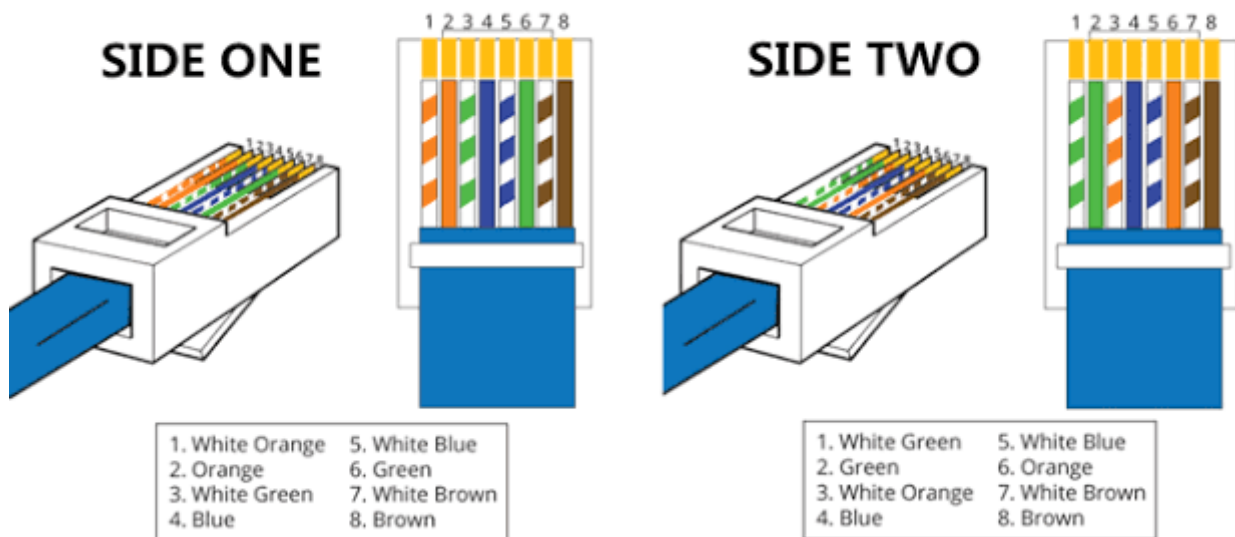
Features of Crossover Cable -

- **Device-to-Device Connection** : Mostly used in direct connectivity between composites of similar types like two computers or two switches.
- **Pin Configuration** : Pins connected on two ends with different arrangements for example T568 A on one end and T568B on the other end.
- **Transmit/Receive Swapping** : Exchange the two ports' T/R signal between the two ends.
- **Specific Purpose** : Typically employed in scenarios where devices must be able to interface without requiring a hub or switch.
- **Colour Code Variation** : In a crossover, some wires at each end have colour-coded sequences different from the other end.



Crossover Cable

CROSSOVER



PRACTICAL 4

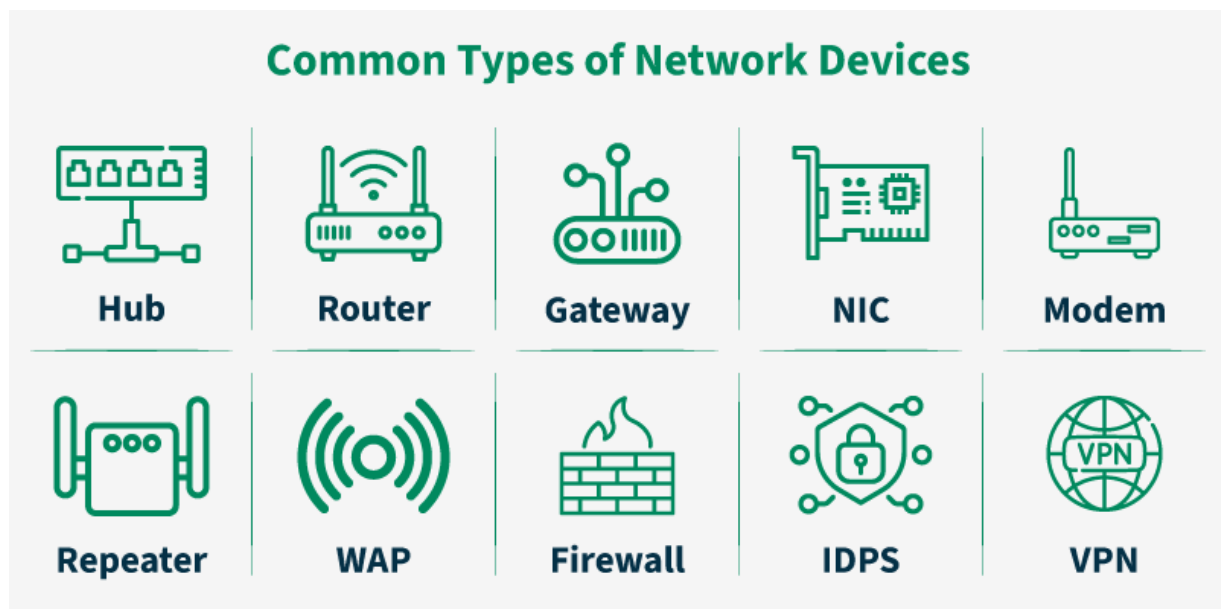
Aim : Study and Test various Network devices available at Department/Institute. (Repeater, Hub, Switch, Bridge, Router and Gateway).

Input :

Network Devices

Network Devices are the physical appliances required for communication and interaction between computers on a computer network.

- Enable communication by transmitting and receiving data between devices.
- Allow devices to connect to networks efficiently and securely.
- Improve network performance by reducing congestion and managing traffic.
- Extend network coverage and solve signal loss or attenuation problems.



Types of Network Devices

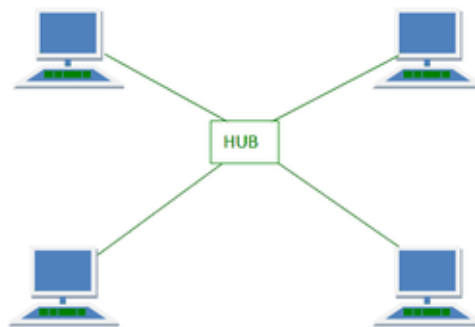
Layer 1 Devices : Physical Layer

These devices deal with raw electrical or optical signals (bits). These devices don't understand IP addresses or MAC addresses.

1. Hub

It is a central connection point for devices in a LAN.

- It takes an incoming signal on one port and blindly broadcasts it to all other ports without knowing the recipient.
- High network traffic collisions and security risks (everyone sees everyone's data).
- It is an obsolete device, replaced by Switches.



Advantages

- **Low cost** : Hubs are very cheap compared to network switches.
- **Easy to use** : They need no setup or special software.
- **Media flexibility** : They can connect different cable types like twisted pair or fiber optic.

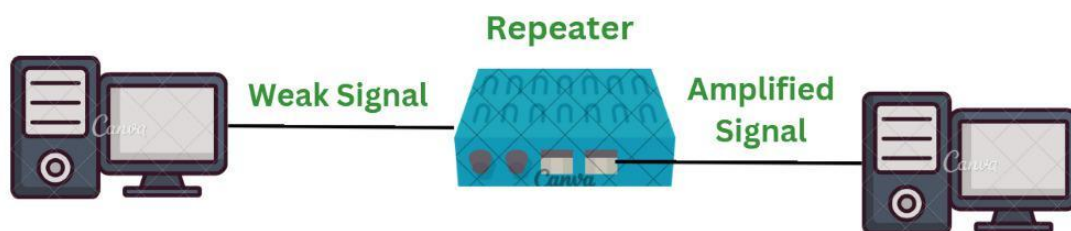
Disadvantages

- **High traffic** : They broadcast data to all ports instead of the target device.
- **Shared speed** : Total bandwidth is shared among all connected devices.
- **Low security** : Any connected device can see all network data.
- **Half-duplex only** : Devices cannot send and receive data at the same time.

2. Repeater

It regenerates signals to extend the range of a network.

- It receives a weak signal (due to attenuation over long cables), amplifies it, and retransmits it.
- Extending WiFi range or Ethernet cables beyond 100 meters.



Advantages

- **Extends range** : Overcomes signal loss (attenuation) over long distances.
- **Cost-effective** : Inexpensive compared to routers or switches.
- **Easy setup** : Simple to install and requires no complex software configuration.
- **No processing overhead** : Works transparently on raw bits without slowing down data via heavy processing.

Disadvantages

- **Amplifies noise** : Boosts background noise right along with the actual signal.
- **No traffic filtering** : Cannot manage, route, or filter data packets.
- **Shared bandwidth** : Does not segment network traffic, which leaves all connected devices in a single collision domain.
- **Limited quantity** : Too many repeaters added in a row can cause signal delays and packet collisions.

Layer 2 Devices: Data Link Layer

These devices work with MAC Addresses (Physical Addresses). They are smarter than hubs.

3. Switch

A switch is a high-speed networking device that connects devices (computers, printers, servers) within a Local Area Network (LAN),

- Unlike hub, switch learns the MAC address of every connected device.
- It sends data only to the specific port where the destination device is connected.
- Reduces collisions and improves security and speed.



Advantages

- **More Speed** : Each port gets full, dedicated bandwidth. Devices do not share speed.
- **Fewer Collisions** : Direct paths mean fewer data crashes on the network.
- **Better Security** : Data goes only to the intended device. It is harder to snoop.
- **Grows Easily** : You can add more ports and devices as needed.

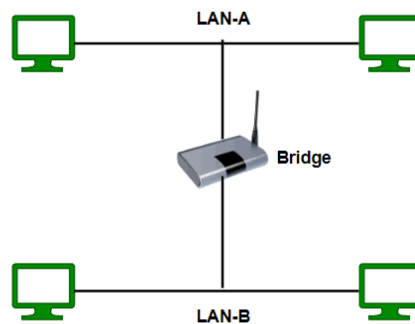
Disadvantages

- **Higher Cost** : Switches cost more than simple hubs.
- **Setup Needed** : Managed types need a person who knows network rules to set them up.
- **Hard to Trace** : Finding a bad link or fault takes more time in big setups.
- **Limited Traffic Control** : They do not handle different subnets or internet routing like routers do.

4. Bridge

This connects two separate LAN segments to make them appear as one.

- Filters traffic based on MAC addresses to keep local traffic local, reducing congestion.
- Largely replaced by Switches (which are essentially multi-port bridges).



Advantages

- **Reduces traffic** : Breaks big networks into smaller parts to stop extra data flow.
- **Fewer collisions** : Creates separate collision domains to give devices faster speeds.
- **Easy setup** : Simple to install without extra complex software.
- **Extends network** : Acts like a repeater to make the network longer.

Disadvantages

- **Slower than switches** : Slower at processing data than modern network switches.
- **No high security** : Does not have advanced security tools like firewalls.
- **Broadcast issues** : Can pass broadcast messages through, which may slow down performance.
- **Limited use** : Works only on local area networks (LANs), not wide area networks (WANs).

Layer 3 Devices: Network Layer

These devices work with IP Addresses (Logical Addresses) and connect different networks together.

5. Router

It is a networking device that connects multiple networks and directs data packets between them using IP addresses.

- It uses IP Addresses to determine the best path for data packets to travel.
- It maintains a Routing Table to make these decisions.
- Routers stop broadcast traffic, isolating networks from each other.



Advantages

- **Internet Sharing** : Allows many devices to share one internet connection.
- **Path Selection** : Finds the fastest and best route for data packets.
- **Network Security** : Protects internal devices using Network Address Translation (NAT) and firewalls.
- **Traffic Control** : Manages and separates network traffic to reduce congestion.

Disadvantages

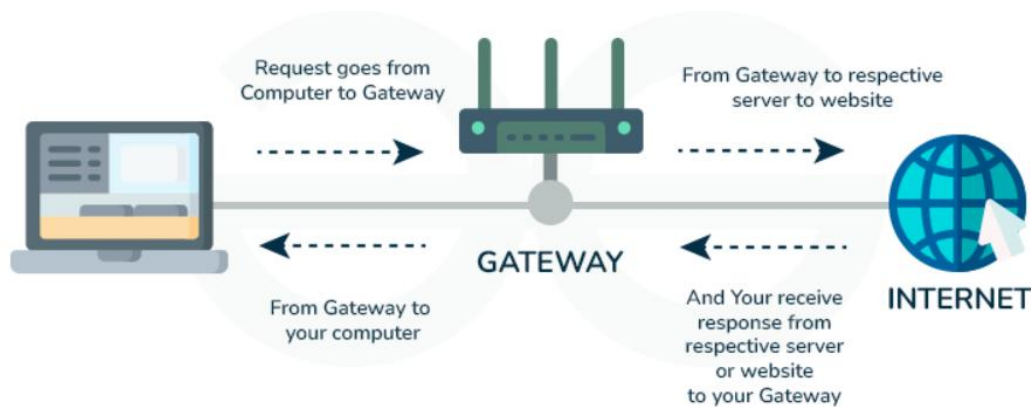
- **Higher Cost** : Advanced or enterprise models cost more than basic switches or hubs.
- **Range Limits** : Wireless signals weaken over distance and through physical walls.
- **Setup Complexity** : Configuration requires basic technical knowledge.
- **Single Point of Failure** : If the main router stops working, connected devices lose internet access.

Layer 4-7 Devices: Advanced Processing

6. Gateway

It is a hardware device or software node that acts as an entry/exit point between two distinct networks using different protocols, translating data to ensure compatibility.

- It can convert protocols, data formats, or architectures (e.g., connecting a TCP/IP network to a legacy Mainframe network).
- An Internet Gateway translates your private LAN requests into public internet requests



Advantages

- **Protocol conversion** : Translates data from one network architecture to another.
- **Network security** : Filters traffic and blocks external threats.
- **Wide connectivity** : Links completely different systems like LANs to WANs or the internet.
- **Centralized control** : Manages data routing efficiently.

Disadvantages

- **High cost** : Expensive to install and maintain.
- **Complex configuration** : Hard to set up and manage.
- **Slower performance** : Protocol translation can add processing delays.
- **Single point of failure** : If the gateway fails, connected networks lose communication.

PRACTICAL 5

Aim : Determine whether following IPv4 address are valid or invalid. If valid IPv4 address then find class, Network and Host ID of an IPv4 address. If invalid IPv4 address then write reason for the same.

- a) 298.27.5.6
- b) 1.4.5.5
- c) 11100010.23.14.67
- d) 192.226.12.11.9

Input :

Sr. No.	Address	Valid / Invalid	Class	Network ID	Host ID	Reason
a)	298.27.5.6	Invalid	—	—	—	First octet 298 > 255, so it is not a valid IPv4 address.
b)	1.4.5.5	Valid	Class A	1	4.5.5	All 4 octets are in the range 0–255.
c)	11100010.23.14.67	Invalid	—	—	—	First octet 11100010 is not a valid decimal octet. If it was intended as binary, it would need to be converted to decimal first.
d)	192.226.12.11.9	Invalid	—	—	—	Contains 5 octets; IPv4 addresses must contain exactly 4 octets.

Explanation of the valid address

b) 1.4.5.5

- First octet = 1
- Since the first octet is 1–126, it belongs to Class A.
- Default Class A structure: Network.Host.Host.Host
- Therefore:
 - Network ID = 1
 - Host ID = 4.5.5

PRACTICAL 6

Aim : Determine Class and Network Address for given IPv4 address and subnet mask.

IPv4 address	Subnet Mask	Class	Subnet
172.16.2.10	255.255.255.0		
10.6.24.20	255.255.240.0		
10.30.36.12	255.255.255.0		

Input :

IPv4 address	Subnet Mask	Class	Subnet
172.16.2.10	255.255.255.0	B	172.16.2.0
10.6.24.20	255.255.240.0	A	10.6.16.0
10.30.36.12	255.255.255.0	A	10.30.36.0

1. 172.16.2.10 / 255.255.255.0

- First octet = 172 - Class B
- Mask = /24
- Network = 172.16.2.0

2. 10.6.24.20 / 255.255.240.0

- First octet = 10 - Class A
- Third-octet mask = 240
- Block size = 256-240 = 16
- 24 falls in 16-31 range
- Network = 10.6.16.0

3. 10.30.36.12 / 255.255.255.0

- First octet = 10 - Class A
- Mask = /24
- Network = 10.30.36.0

PRACTICAL 7

Aim : Subnet the IP address 216.21.5.0 / 24 into 30 hosts in each subnet.

Input :

1. Given information

- IP address: 216.21.5.0
- Original prefix: /24
- Original subnet mask: 255.255.255.0
- Required hosts per subnet: 30 usable hosts

We need to find the new subnet mask and list all the resulting subnets.

2. Determine how many host bits are required

The formula for the number of usable hosts in a subnet is:

$$2^h - 2$$

where h = number of host bits.

We need at least 30 hosts.

Try 4 host bits:

$$2^4 - 2 = 16 - 2 = 14$$

Only 14 hosts — not enough.

Try 5 host bits:

$$2^5 - 2 = 32 - 2 = 30$$

Exactly 30 usable hosts.

Therefore, we need 5 host bits.

3. Find the new prefix length

The original network is /24.

IPv4 has 32 bits:

$$32 - 24 = 8$$

So the original /24 network has 8 host bits.

We need 5 host bits for each subnet. Therefore, we borrow:

$$8 - 5 = 3 \text{ bits}$$

from the host portion.

So:

$$24 + 3 = \boxed{/27}$$

New subnet prefix

$$\boxed{/27}$$

4. Find the subnet mask

A /27 means the first 27 bits are network bits.

Binary subnet mask:

11111111.11111111.11111111.11100000

Convert each octet to decimal:

$$11111111 = 255$$

$$11111111 = 255$$

$$11111111 = 255$$

$$11100000 = 224$$

Therefore:

$$\boxed{255.255.255.224}$$

So the answer is:

Subnet mask = 255.255.255.224 (/27)

5. Determine the number of subnets

We borrowed 3 bits from the original host portion.

Number of subnets:

$$2^3 = 8$$

Therefore, we get:

8 subnets

Each subnet has:

$$2^5 = 32$$

total IP addresses.

Of these:

- 1 = Network address
- 30 = Usable host addresses
- 1 = Broadcast address

Therefore:

$$32 - 2 = 30 \text{ usable hosts}$$

6. Find the block size

The interesting octet is the fourth octet.

Our subnet mask is:

255.255.255.224

Block size is:

$$256 - 224 = 32$$

So the networks increase by 32 in the fourth octet.

Starting from 0:

0

32

64

96

128

160

192

224

These are our eight network addresses.

7. List all subnets

Subnet 1

Network address:

216.21.5.0

The next network starts at .32, so usable hosts are:

216.21.5.1 – 216.21.5.30

Broadcast:

216.21.5.31

Therefore:

Item	Address
Network	216.21.5.0/27
First host	216.21.5.1
Last host	216.21.5.30
Broadcast	216.21.5.31

Subnet 2

Network address:

216.21.5.32

Usable hosts:

216.21.5.33 – 216.21.5.62

Broadcast:

216.21.5.63

Item	Address
Network	216.21.5.32/27
First host	216.21.5.33
Last host	216.21.5.62
Broadcast	216.21.5.63

Subnet 3

Network address:

216.21.5.64

Usable hosts:

216.21.5.65 – 216.21.5.94

Broadcast:

216.21.5.95

Item	Address
Network	216.21.5.64/27
First host	216.21.5.65
Last host	216.21.5.94
Broadcast	216.21.5.95

Subnet 4

Network address:

216.21.5.96

Usable hosts:

216.21.5.97 – 216.21.5.126

Broadcast:

216.21.5.127

Item	Address
Network	216.21.5.96/27
First host	216.21.5.97
Last host	216.21.5.126
Broadcast	216.21.5.127

Subnet 5

Network address:

216.21.5.128

Usable hosts:

216.21.5.129 – 216.21.5.158

Broadcast:

216.21.5.159

Item	Address
Network	216.21.5.128/27
First host	216.21.5.129
Last host	216.21.5.158
Broadcast	216.21.5.159

Subnet 6

Network address:

216.21.5.160

Usable hosts:

216.21.5.161 – 216.21.5.190

Broadcast:

216.21.5.191

Item	Address
Network	216.21.5.160/27
First host	216.21.5.161
Last host	216.21.5.190
Broadcast	216.21.5.191

Subnet 7

Network address:

216.21.5.192

Usable hosts:

216.21.5.193 – 216.21.5.222

Broadcast:

216.21.5.223

Item	Address
Network	216.21.5.192/27
First host	216.21.5.193
Last host	216.21.5.222
Broadcast	216.21.5.223

Subnet 8

Network address:

216.21.5.224

Usable hosts:

216.21.5.225 – 216.21.5.254

Broadcast:

216.21.5.255

Item	Address
Network	216.21.5.224/27
First host	216.21.5.225
Last host	216.21.5.254
Broadcast	216.21.5.255

8. Complete answer in one table

	Network Address	First Host	Last Host	Broadcast
1	216.21.5.0/27	216.21.5.1	216.21.5.30	216.21.5.31
2	216.21.5.32/27	216.21.5.33	216.21.5.62	216.21.5.63
3	216.21.5.64/27	216.21.5.65	216.21.5.94	216.21.5.95
4	216.21.5.96/27	216.21.5.97	216.21.5.126	216.21.5.127
5	216.21.5.128/27	216.21.5.129	216.21.5.158	216.21.5.159
6	216.21.5.160/27	216.21.5.161	216.21.5.190	216.21.5.191
7	216.21.5.192/27	216.21.5.193	216.21.5.222	216.21.5.223
8	216.21.5.224/27	216.21.5.225	216.21.5.254	216.21.5.255

Final result

Original network: 216.21.5.0/24

New subnet: /27

Subnet mask: 255.255.255.224

Number of subnets: 8

Total addresses per subnet: 32

Usable hosts per subnet: 30

Quick trick to remember

For a /27:

Mask = 255.255.255.224 → Block size = 32

Then simply count by 32:

0, 32, 64, 96, 128, 160, 192, 224

For each one:

Network → Network + 1 through Network + 30 → Network + 31 (Broadcast)

PRACTICAL 8

Aim : Identify valid IPv6 addresses and if invalid IPv6 address then write reason for the same:

2001: db8: 3333: 4444: 5555: 6666: 7777: 8888
2001:0db8:0001:0000: 0000:0ab9:C0A8:0102
225.1.4.2
::
fe80:2030:31:24

Input :

Absolutely. Let's go through each address carefully, using the rules for IPv6.

IPv6 basics

An IPv6 address is 128 bits long and is normally written as 8 groups, called hextets, separated by colons (:).

Example :

2001:0db8:0000:0000:0000:ff00:0042:8329

Each hextet :

- Contains 1 to 4 hexadecimal digits.
- Hexadecimal digits can be :
 - 0–9
 - a–f
 - A–F
- There must normally be 8 hextets.
- :: can be used once to represent one or more groups of zeros.
- Spaces are not allowed inside an IPv6 address.

1. 2001: db8: 3333: 4444: 5555: 6666: 7777: 8888

Answer : Invalid

Let's examine it :

2001: db8: 3333: 4444: 5555: 6666: 7777: 8888

There are apparently 8 groups :

1. 2001
2. db8
3. 3333
4. 4444
5. 5555
6. 6666
7. 7777
8. 8888

The number of groups is correct, and all the characters are hexadecimal.

However, there are spaces after the colons.

For example :

2001: db8

^

The correct IPv6 notation cannot contain these spaces.

Correct form

Remove the spaces :

2001:db8:3333:4444:5555:6666:7777:8888

This is a valid IPv6 address.

Conclusion

Given address : Invalid

Reason : Spaces are present.

Without spaces : Valid

2. 2001:0db8:0001:0000:0000:0ab9:C0A8:0102

Answer : Valid

Let's divide it into its 8 groups :

2001 : 0db8 : 0001 : 0000 : 0000 : 0ab9 : C0A8 : 0102

Now check each group :

Group	Value	Valid?
1	2001	✓
2	0db8	✓
3	0001	✓
4	0000	✓
5	0000	✓
6	0ab9	✓
7	C0A8	✓
8	0102	✓

Every group contains 4 hexadecimal digits or fewer.

Notice :

C0A8 might look unusual because it contains letters, but that's completely fine.

In hexadecimal :

C = 12

0 = 0

A = 10

8 = 8

So C0A8 is a perfectly valid hexadecimal hextet.

There are also exactly 8 groups, so the address represents 128 bits.

Conclusion

2001:0db8:0001:0000:0000:0ab9:C0A8:0102 is a valid IPv6 address.

3. 225.1.4.2

Answer : Invalid as an IPv6 address

This address uses dots (.) :

225.1.4.2

IPv4 addresses are written using four decimal numbers separated by dots.

For example :

192.168.1.1

The given address :

225.1.4.2 follows the IPv4 format.

IPv6, on the other hand, uses hexadecimal groups separated by colons :

2001:db8::1

Therefore :

225.1.4.2 is not an IPv6 address.

Important distinction

It is actually a valid IPv4 address because:

- It has 4 decimal parts.
- Each part is between 0 and 255.
- The parts are separated by dots.

So:

As IPv4 : Valid

As IPv6 : Invalid

4. ::

Answer : Valid

This is an important special case.

IPv6 allows a special notation called zero compression.

The notation :

:: represents one or more groups of zeros.

For example:

0:0:0:0:0:0:0:0

can be shortened to ::

Therefore:

:: represents the IPv6 address :

0000:0000:0000:0000:0000:0000:0000:0000

This is called the unspecified IPv6 address.

Why is :: allowed?

Normally IPv6 requires 8 groups :

xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx

But :: can replace consecutive zero groups.

For example :

2001:0db8:0000:0000:0000:0000:0000:0001

can become : 2001:db8::1

Here, :: replaces five groups of zeros.

Important rule

:: can appear at most once in an IPv6 address.

For example :

2001::abcd::1234 is invalid because it contains two ::.

Conclusion

:: is a valid IPv6 address.

5. fe80:2030:31:24

Answer : Invalid

Let's count the groups :

fe80 : 2030 : 31 : 24

There are only 4 groups :

1. fe80
2. 2030
3. 31
4. 24

A normal IPv6 address requires 8 groups.

For example :

fe80:0000:0000:0000:0000:0000:0001

The given address has no :: to indicate that some zero groups have been omitted.

Could it be shortened?

Yes, but it would need ::.

For example, if the intended address were :

fe80:2030:31:24:0000:0000:0000:0000

it could be written as : fe80:2030:31:24::

That would be valid.

But :

fe80:2030:31:24 by itself does not contain enough information to represent all 128 bits.

Conclusion

Given address : Invalid

Reason : Only 4 hexets are present and there is no :: compression.

Final answer in exam format :

No.	Given Address	Valid / Invalid	Reason
1	2001: db8: 3333: 4444: 5555: 6666: 7777: 8888	Invalid	Spaces are present. IPv6 addresses cannot contain spaces.
2	2001:0db8:0001:0000:0000:0ab9:C0A8:0102	Valid	Contains exactly 8 valid hexadecimal hextets.
3	225.1.4.2	Invalid IPv6	It is an IPv4 address, using decimal numbers and dots.
4	::	Valid	:: is valid IPv6 zero compression and represents groups of zeros.
5	fe80:2030:31:24	Invalid	Only 4 hextets are present;

PRACTICAL 9

Aim : Study of firewall in providing network security.

Input :

Introduction :

A firewall is a network security system that monitors and controls incoming and outgoing network traffic based on predefined security rules. It acts as a barrier between a trusted internal network and an untrusted network, such as the Internet.

Objectives of a Firewall :

- To prevent unauthorized access to a network.
- To protect computers and network resources from malicious traffic.
- To monitor and control network communication.
- To block suspicious or unwanted connections.
- To enforce an organization's security policies.

How a Firewall Provides Network Security :

A firewall examines network packets and decides whether to allow, reject, or block them according to configured rules. For example, it can allow web traffic through HTTP/HTTPS while blocking unauthorized ports or connections.

The basic process is:

Internet → Firewall → Internal Network

The firewall checks traffic before it reaches protected systems, thereby reducing the risk of unauthorized access and attacks.

Types of Firewalls :

1. **Packet-Filtering Firewall** – Examines packet information such as source/destination IP addresses, ports, and protocols.
2. **Stateful Firewall** – Tracks active connections and makes decisions based on the state of network sessions.
3. **Proxy Firewall** – Acts as an intermediary between users and external networks.
4. **Next-Generation Firewall (NGFW)** – Provides advanced inspection, application control, intrusion prevention, and other security capabilities.
5. **Host-Based Firewall** – Runs directly on an individual computer or server.

Advantages :

- Protects against unauthorized network access.
- Controls network traffic.
- Helps prevent certain types of malware and attacks.
- Provides logging and monitoring of network activity.
- Supports enforcement of security policies.
- Can help isolate sensitive parts of a network.

Limitations :

- A firewall cannot protect against every type of attack.
- It cannot fully prevent attacks caused by compromised or careless users.
- Incorrect firewall configuration can create security vulnerabilities.
- Encrypted traffic may require additional security mechanisms for effective inspection.
- Firewalls should be used together with antivirus, intrusion detection/prevention, authentication, and other security measures.

Conclusion :

A firewall is an important component of network security because it creates a controlled boundary between trusted and untrusted networks. By filtering and monitoring traffic according to security rules, it helps protect systems and data from unauthorized access. However, a firewall is not a complete security solution and should be combined with other security controls for effective protection.

PRACTICAL 10

Aim : Run basic utilities and network commands : ipconfig, ping, tracert, netstat, pathping, route.

Input :

PRACTICAL 11

Aim : List different adapters in any computer/ laptop and configure network connection in your institute/ premises LAN.

Input :